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Noninvasive Worker Safety Monitoring and Augmented Reality Feedback for Real-Time Intervention

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Abstract

To more effectively address the wide range of safety risks faced by construction workers on job sites, machine learning (ML)–based computer vision and augmented reality (AR) technologies are increasingly being employed to enhance efficiency, safety, and productivity. However, current AR construction safety tools only provide passive information for the user to then decide how to use that information. This study leverages advanced computer vision coupled with AR to work with site managers and on-site workers to make operational safety decisions using real-time, visual information of potential hazards. A YOLOv11 model trained to detect the presence or lack of personal protective equipment was developed and tested by creating a local ML computing environment using camera feeds. The detection results were compiled and displayed in real time on a web-based interface developed with Hypertext Preprocessor and on a Microsoft HoloLens 2 heads-up display. The system was successfully field-tested on a construction site.

Keywords: AR; computer vision; construction safety; detection; PPE; SSHO; YOLO

1. Introduction

Despite the common recognition of operational risk management as essential in the construction industry, construction workers face a wide array of poorly mitigated risks on the jobsite. The greatest of these risks include falls from scaffolding, ladders, or rooftops, being struck by objects or machinery, exposure to hazardous substances, electrical hazards, and accidents involving heavy equipment. Per the 2023 National Census of Fatal Occupational Injuries by the U.S. Bureau of Labor Statistics (BLS), there were 1075 construction worker fatalities in the U.S., which represented 20.3% of all worker fatalities across industries in the U.S. in 2023 [1]. The U.S. Occupational Safety and Health Administration (OSHA) reported that falls accounted for 39.2% of construction worker deaths in 2023. Moreover, struck-by incidents were responsible for 24.9% of construction worker fatalities, which represented poor risk management of vehicles and heavy equipment like cranes, forklifts, and bulldozers [2]. A major risk management practice to help mitigate hazards associated with struck-by incidents is the use of personal protective equipment (PPE), which includes, but is not limited to, hard hats and high-visibility, reflective safety vests.

There are three roles that must be filled in most federal projects: the superintendent, the quality control manager (QCM), and the SSHO. The superintendent is primarily responsible for on-site production, management of materials, and on-site management of

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subcontractors. The quality control manager uses a three-phase process of quality control, which includes a preparatory phase review of project plans and specifications, an initial phase verification that the crew is following proper procedures, and a follow-up phase of documenting adherence to or departure from plans and specifications. While the superintendent and the QCM's function can help identify safety concerns, their role is heavily weighted in coordination with different crews and subject matter experts, meetings, and paperwork. The site safety and health officer's (SSHO) primary responsibility is to ensure that the crew adheres to published safety standards, including EM-385-1-1 and applicable OSHA regulations. The SSHO carries out this responsibility by conducting routine safety training for the crew, safety planning using the Operational Risk Management process, and oversight of the crew while they are carrying out their work [3].

SSHOs spend most of their time in the field, roaming through different areas of the construction site and observing different crews as they accomplish their work. One of the biggest and most easily correctable safety issues on a construction site is the presence or absence of required PPE. One main challenge for site management is that crews will wear the required PPE while the SSHO is watching and then take the PPE off when management walks away. In addition, they typically cannot supervise the entire construction at once, especially if there are multiple crews working in multiple areas. Jobsite management would benefit from real-time, active safety hazard identification tools to help site management mitigate hazards across the whole construction site at once.

Computer vision has been adopted to monitor people and their behavior. Studies have focused on collecting detection data on existing camera feeds to collect safety data. Draughon et al. [4] proposed a computer vision framework utilizing Mask R-CNN (recursive convolutional neural network) and multi-target tracking for the use of face masks on the Detroit Riverwalk (i.e., an outdoor public setting) during the COVID-19 pandemic. The study focused on identifying persons and whether they were wearing a mask. The study achieved 96% accuracy despite citing difficulties with low-resolution images. Draughon extended this work, focusing on Human Activity Recognition (HAR) by implementing the YOLO-v3 (You Only Look Once) model on camera feeds and wearable GPS trackers for monitoring human behavior [5]. Nath et al. [6] employed YOLO-v3 for real-time detection of construction objects, demonstrating effectiveness in diverse lighting conditions and highlighting the potential for real-time monitoring of worker activity. Lee et al. [7] used convolutional neural network (CNN) based models for fall risk assessment and PPE compliance on camera feeds. Synthetic datasets enhanced their model's robustness, providing valuable insights into simulating high-risk scenarios for worker safety. Alateeq et al. [8] implemented YOLO-v5 for identifying hazards and PPE compliance on closed-circuit television (CCTV) under varied environmental conditions, incorporating weather data for proactive safety measures. Shad et al. [9] conducted a State-of-the-Art analysis of the integration of augmented reality (AR) with construction technologies to improve construction safety. The study explored the utility of different technologies, and a useful research question proposed was the specific aims of AR applications in the construction safety academic literature.

Regarding worker behavior, in a study interviewing 60 construction workers, Wong et al. [10] found that some common reasons for workers not wearing PPE are the inconvenience of the PPE, physical discomfort, risk perception, and social influence from their coworkers. The study also found that the workers are much more likely to wear PPE when they have been trained and are being supervised.

Recent work has focused on improving PPE detection using more advanced deep learning architectures to improve accuracy, robustness, and real-time capability in construction environments. Wang et al. [11] proposed a detection framework that combined YOLOv10

with transformer backbones. The model demonstrated strong detection performance across multiple PPE categories (i.e., helmets, masks, vests, gloves, and shoes) in complex construction scenes using surveillance and body-worn cameras and improved feature representation compared to traditional CNNs. Liu et al. [12] proposed PEG-YOLOv10m for helmet detection, which incorporates a hybrid attention mechanism to improve detection accuracy in construction environments. This study demonstrated improved performance in small-object detection and low-light and occluded conditions. Jena et al. [13] proposed a helmet detection framework using the Detection Transformer (DETR), which combines a CNN backbone with a transformer encoder–decoder architecture. The transformer-based model improves handling of occlusion, complex spatial relationships, and contextual understanding in detection tasks but results in slower inference compared to lightweight YOLO-based models. Li et al. [14] proposed SD-YOLOv5, an architecturally enhanced YOLOv5 model to improve the detection of small objects, such as helmets, in complex environments. Eum et al. [15] proposed a YOLOv10-based framework with transformer backbones for real-time heavy equipment detection on construction sites, demonstrating improved detection accuracy and multi-scale feature representation.

More recent studies have expanded this work beyond single-task PPE detection toward multimodal safety monitoring, broader noncompliance categories, and improved generalization. Sivanraj et al. [16] developed a real-time construction safety detection framework that integrated YOLOv11s object detection, a vision language model, and natural language processing to detect PPE compliance, hazardous ground openings, safety measures, and environmental conditions, while also generating real-time Telegram alerts and weekly safety reports. Wang et al. [17] proposed a YOLOv10-based multi-scale variant detector for multi-category PPE compliance monitoring, using Omni Dimensional Dynamic Convolution, an Efficient Multi-scale Attention guided Bidirectional Feature Pyramid Network, Minimum Points Distance IoU loss, and a four scale prediction strategy to improve detection of no helmet, no mask, no vest, no glove and no shoe conditions, particularly for small, distant and partially occluded PPE violation cues. Wang [18] also developed a domain-adaptive Faster R-CNN framework for non-PPE identification that used construction site imagery and general context images to address the scarcity of authentic noncompliance examples on regulated jobsites, detecting no helmet, no mask, no glove, no vest, and no shoes conditions while improving generalization across image sources.

Despite these advances, the existing literature remains primarily focused on improving detection accuracy, model robustness, and architecture design. These contributions are necessary for automated construction safety monitoring, but detection performance alone does not resolve how safety information should be delivered to the site safety management team. In many practical construction settings, SSHOs and site managers must divide their attention between the work area, remote crews, and separate monitoring devices. This separation may delay recognition of safety-relevant events, particularly on dynamic or spatially distributed jobsites. Therefore, there is a need for system-level research that integrates real-time computer vision detection with end-user visualization tools capable of delivering safety information directly to the SSHOs and other members of the construction management team in a format that supports timely hazard recognition and intervention.

Conversely, heads-up AR systems can enhance situational awareness by reducing cognitive load and minimizing the need to divide attention between physical environments and external displays [19]. This division of attention can reduce the effectiveness of hazard identification and timely intervention, especially on dynamic and dispersed construction sites. Moreover, construction projects often involve multiple, and sometimes geographically dispersed, active work zones, moving equipment, and limited lines of sight, which

makes it difficult for a single SSHO to maintain continuous awareness using traditional monitoring approaches.

This research developed an integrated hardware–software system that was designed to support real-time safety decision-making on construction sites. The system combined computer vision detection models with AR to project critical safety alerts directly into the SSHO's field of view, enabling continuous monitoring of both nearby activities and remote areas of the jobsite. By using AR as a primary situational awareness tool, the platform enhances the speed and effectiveness of hazard recognition and safety interventions in complex, dynamic, and dispersed construction environments. The system also compiled and displayed potential hazards on a web-based user interface for other construction management personnel, providing a comprehensive, real-time snapshot of safety information across the jobsite. By integrating detection, both AR and web-based visualization, and data management into a unified framework, this work contributes a system-level approach for delivering accurate, real-time, and actionable safety information to the SSHO and construction management team, while demonstrating how such integrated systems can support real-time safety decision-making in large or complex construction sites.

Based on this research gap, the present study was guided by three objectives. The first objective was to evaluate whether a YOLOv11-based object detection model trained on a construction safety dataset could identify workers, PPE compliance states, and selected safety-related objects with sufficient performance for real-time monitoring. The second objective was to verify whether the integrated camera, local database, web interface, and AR display pipeline could transmit and visualize detection results under representative field conditions. The third objective was to assess the feasibility of AR as an information delivery method for SSHOs by presenting safety alerts in a heads-up format that supports situational awareness without having to rely solely on separate external displays.

2. Materials and Methods

2.1. System Overview

The integrated hardware–software system primarily relies on XAMPP for data management while it gathers detection data from both stationary camera feeds and a camera feed from the HoloLens 2 AR headset, using custom Python 3.14.4, PHP 8.5, Unity C#, and HTML scripts to pass and display the detection data. First, a trained You Only Look Once (YOLO) version 11 model is deployed using Python in PyCharm 2024.1.4 to gather detection information from these cameras in parallel using multithreading. The script parses detection output to extract bounding boxes, class labels, and confidence scores for each detection event. This metadata is then passed to a XAMPP MySQL database in JSON format via HTTP POST requests using a PHP script. An additional PHP script responds to HTTP GET requests from the HoloLens and the web-based user interface. A Unity C# script is used to display the detection information on the HoloLens, and HTML coding is used to display the detection information on a web-based user interface. A visualization of this data pipeline is shown in Figure 1 below.

The system was intentionally designed for modularity to reduce latency and enable independent updates of the scripts and/or camera streams without disrupting the entire system. It also uses standard, widely available technologies, such as HTTP POST/GET requests, JSON formatting, and a XAMPP-based local server, to ensure compatibility across different computing platforms. Furthermore, the system's script design supports scalability, allowing new detection classes or additional camera zones to be added with minimal reconfiguration.

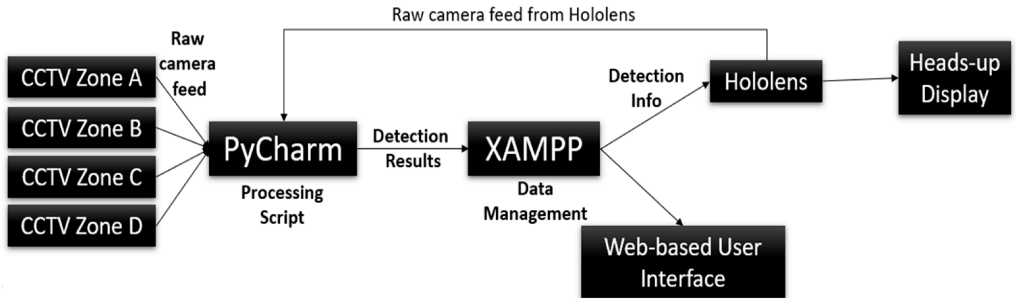


Figure 1. Data pipeline for an integrated hardware–software system.

2.2. Background, Dataset Development, and Model Training

2.2.1. Convolutional Neural Networks

A convolutional neural network is a type of neural network that specializes in processing data that has a grid-like topography. One of the more common examples of this data is an image, whose pixels can be represented numerically in terms of pixel intensity or color channels, forming a matrix. This numerical representation allows the CNN to perform classification, pattern recognition, and object detection. In the first layer of the neural network, small, square matrices called filters (or kernels) specify patterns for which to look [20]. As a filter moves across the grid convolutionally, or in a spatially incremental manner, it evaluates groups of pixels and gives each group a numeric score based on how closely it compares to the filter. After every group of pixels is mapped, the result is an array of numbers that indicate how closely the image matches the filter [21]. This resulting output is called a feature map [20]. Then, in a process called pooling, regions (or neighborhoods) of the feature map are statically summarized compared to nearby outputs. In the subsequent layers of the neural network, the filters become more abstract and can start to recognize complex shapes based on the resulting numeric scores [20,21]. Eventually, bounding boxes and confidence scores are formed around object detections.

2.2.2. You Only Look Once (YOLO) Detection Model

In the YOLO model, a single neural network predicts bounding boxes and class probabilities directly from full images in one evaluation. This allows the model to train on full images and directly optimizes detection performance. This attribute is especially useful in detecting objects in live video streams. In its inception in 2015, YOLO outperformed models like region-based convolutional neural networks (R-CNN) and deformable parts model (DPM) by avoiding generalization errors when it processed an entire image at once. However, this original YOLO model lagged behind State-of-the-Art detection systems in accuracy [22].

The YOLO model has had several new versions, each with improved accuracy, speed, and efficiency, including YOLOv11 [23]. YOLOv11 showcases a more efficient architecture with C3K2 blocks, Spatial Pyramid Pooling Fast (SPPF), and C3PSA, an advanced attention mechanism. These attributes enhance small object detection and improve accuracy while keeping YOLO's trusted real-time inference speed [23]. While newer versions of YOLO, such as YOLO26, have been introduced, YOLOv11 was chosen for its balance between detection accuracy and real-time inference performance, as well as its robustness in detecting small or occluded objects on complex construction backgrounds [24]. Because the objective of this study was to evaluate the integration of computer vision detection with local database communication, web-based visualization, and AR feedback, a controlled benchmark comparison among YOLO versions was outside the scope of this proof-of-concept study.

2.2.3. Algorithm Training and Validation

A custom dataset, named WorkerSafety25, was created on Roboflow.com by combining several publicly available image and label sets on this platform [25,26]. Public datasets were selected based on the fidelity of the annotations and the balancing of the number of annotations for each class. These datasets were merged into a unified format on Roboflow using consistent class definitions and labeling structure. Annotations were manually validated and corrected on the Roboflow platform, including verification of accurate bounding boxes and class label consistency. Low-quality and duplicate images were removed. The resulting dataset contains 8316 unique images annotated with eight object classes. These classes, along with the number of annotations for each, are shown in Table 1 below.

Table 1. Dataset Classes and Annotations.

Class	Number of Annotations
Person	4199
Safety vest	4629
No safety vest	2583
Hardhat	2858
No hardhat	2839
Gloves on	3625
No glove	2121
Ladder	2309

The selected classes enable the detection of workers and assessment of whether they are wearing commonly required PPE, such as safety vests, hard hats, or gloves. In addition, a "ladder" class was included, as the presence of a ladder may indicate higher-risk work activities that warrant direct observation by the SSHO. The dataset was curated to include a balanced number of annotations for each class to help reduce generalization error, or the expected error when the model is processing new inputs [20]. In other words, balancing the dataset was intended to improve the model's performance on real-world images, such as those captured from live jobsite camera feeds.

The Roboflow web application was used to split the dataset, preprocess the images, and perform data augmentation before model training. First, the dataset was split into approximately 6500 training images, 1200 validation images, and 660 test images before augmentation. Preprocessing was then applied by auto-orienting the images and resizing them to 640 $\times$ 640 pixels. Data augmentation was applied only to the training images during dataset version generation. The augmentation settings produced three outputs per training example and included image rotation between -15° and $+15^{\circ}$ to simulate camera tilt and viewpoint variation, as well as brightness adjustment between -15% and $+15\%$ to mimic shadows, overexposure, and time-of-day differences. Larger rotations were avoided to prevent unrealistic object orientations, and the brightness range was selected to reflect moderate environmental variation typical of construction sites without washing out features or degrading label visibility. The final exported dataset contained 21,310 images, including 19,494 training images, 1156 validation images, and 660 test images. Because the dataset was split before augmentation, and augmentation was only applied to the training images, augmented copies of the same source image were not distributed across the training, validation, and test subsets.

The dataset was then used to perform transfer learning using the YOLOv11 model in the Ultralytics YOLO framework in Google Colab. The model was initialized with pre-trained YOLOv11 weights, and training was performed for 25 epochs to provide sufficient convergence during transfer learning while minimizing the risk of overfitting. A batch size of 8 was selected because of hardware limitations and is consistent with common training practices for models with high-resolution images. Training convergence was assessed using the learning curves generated during model development, including training and validation losses, precision, recall, Mean Average Precision (mAP), and mAP averaged across multiple Intersection over Union (IoU) thresholds from 0.50 to 0.95. As shown in Figure 2, the loss curves generally decreased over the training period, while the validation performance metrics stabilized toward the final epochs. Standard defaults of 0.5 confidence and 0.45 IoU were used to balance precision and recall and to provide a reasonable balance between localization accuracy and detection sensitivity. The confidence score indicates the model's certainty that a detected object belongs to a particular class, and IoU measures how well the predicted bounding box overlaps with the ground-truth bounding box [27].

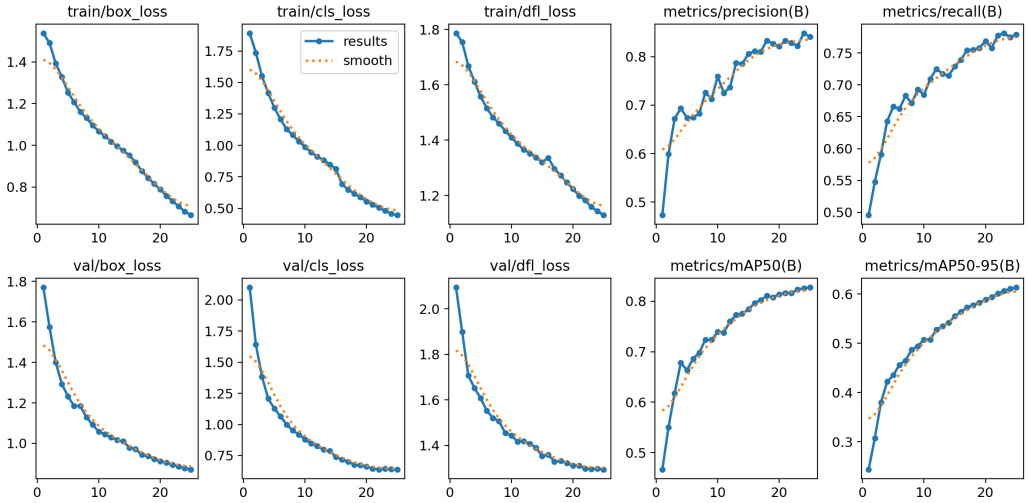


Figure 2. YOLOv11 training and validation learning curves over 25 epochs.

2.3. Hardware Development and Field Integration

A rugged, user-friendly, and quickly deployable hardware system was developed to ensure a low barrier to entry for implementing the hazard detection system on typical construction sites, even for users with varying levels of technical expertise. The system's modular design allows components to be swapped, added, or removed as needed. The basic hardware required for field deployment includes a Wi-Fi router with a power source, a computer, one or more mounted cameras capable of RTSP or webcam mode, and an AR device such as the Microsoft HoloLens 2. While an internet connection to the Wi-Fi router is optional, it enables functionality of the Leaflet API for map-based visualization and allows remote users to access the web-based user interface.

Field validation was conducted as a proof-of-concept deployment on an active outdoor construction site to verify that the integrated camera, database, web interface, and HoloLens 2 display pipeline functioned under representative site conditions. The system was tested using a Wi-Fi router without an internet connection, a laptop computer, two camera feeds, and a Microsoft HoloLens 2 headset. The two camera feeds consisted of the laptop webcam and an Insta360 ONE X2 360-degree camera. The site included seven workers during the test. The validation procedure focused on confirming zone occupancy detection, PPE detection, and successful data transfer from the Python detection script to the local MySQL database and then to the web-based interface and the HoloLens 2 heads-up display. Because

this deployment was intended to verify system integration and field feasibility, it did not include formal measurement of incident reduction, SSHO response time, or statistically sampled false alarm rates.

3. Results

3.1. Worker and PPE Validation Results

The F1–Confidence Curve, the Normalized Confusion Matrix, the Precision–Confidence Curve, and the Precision–Recall Curve for the YOLOv11 training with the WorkerSafety25 dataset are analyzed in this section to provide a comprehensive set of evaluation metrics for the resulting model.

The F1–Confidence Curve in Figure 3 provides a balanced assessment of the model's performance based on the F1 Score, the harmonic mean of precision and recall, across various confidence thresholds [28]. In other words, it reflects a trade-off between True Positives (*TP*), False Positives (*FP*), and False Negatives (*FN*). Goodfellow et al. [20] referred to the F1 score as the F-score, given by Equation (1) below:

$$F1 = \frac{2pr}{p + r} \tag{1}$$

where p is the precision, defined by the following:

$$p = \frac{TP}{TP + FP} \tag{2}$$

and r is the recall, defined by the following:

$$r = \frac{TP}{TP + FN} \tag{3}$$

Combining Equations (1)–(3), the equation for the F1 score given by scikit-learn is shown below [29]:

$$F1 = \frac{2 * TP}{2 * TP + FP + FN} \tag{4}$$

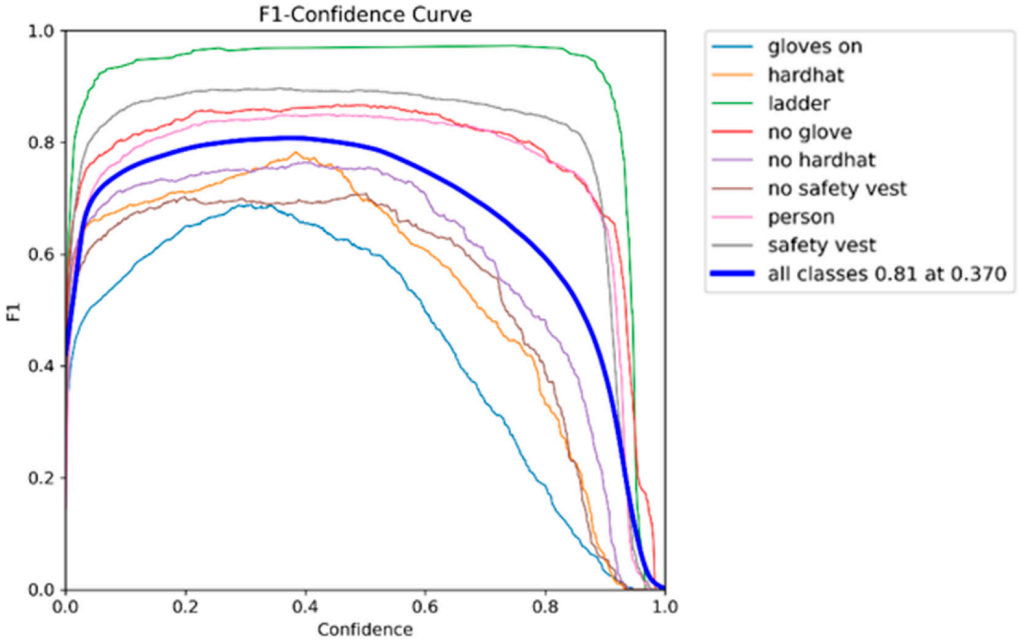


Figure 3. F1–Confidence Curve.

For the newly trained model, Figure 3 shows that the highest average F1 score of 0.81 was achieved at a confidence threshold of 0.37. Class-specific curves that approximate the square shape of the chart's boundary reflect strong F1 performance across a wide range of confidence thresholds, which is an ideal outcome. The top-performing classes are "ladder," "hardhat," and "no glove," each demonstrating high reliability in detection. The "no safety vest" and "person" classes exhibit moderate performance, while the "gloves on" class has the lowest F1 score, indicating an area for potential improvement in future model iterations. Nevertheless, the overall results are acceptable given that "no glove" detections are more critical for hazard identification than confirming "gloves on".

Figure 4 shows the normalized confusion matrix for the trained model. This chart summarizes the model's performance by comparing predicted classifications against the ground-truth labels for images in the validation set [28]. A value of 1 along the diagonal in Figure 4 would indicate perfect classification accuracy for each class. Off-diagonal values represent instances where the model confused one class for another or misclassified background elements as objects.

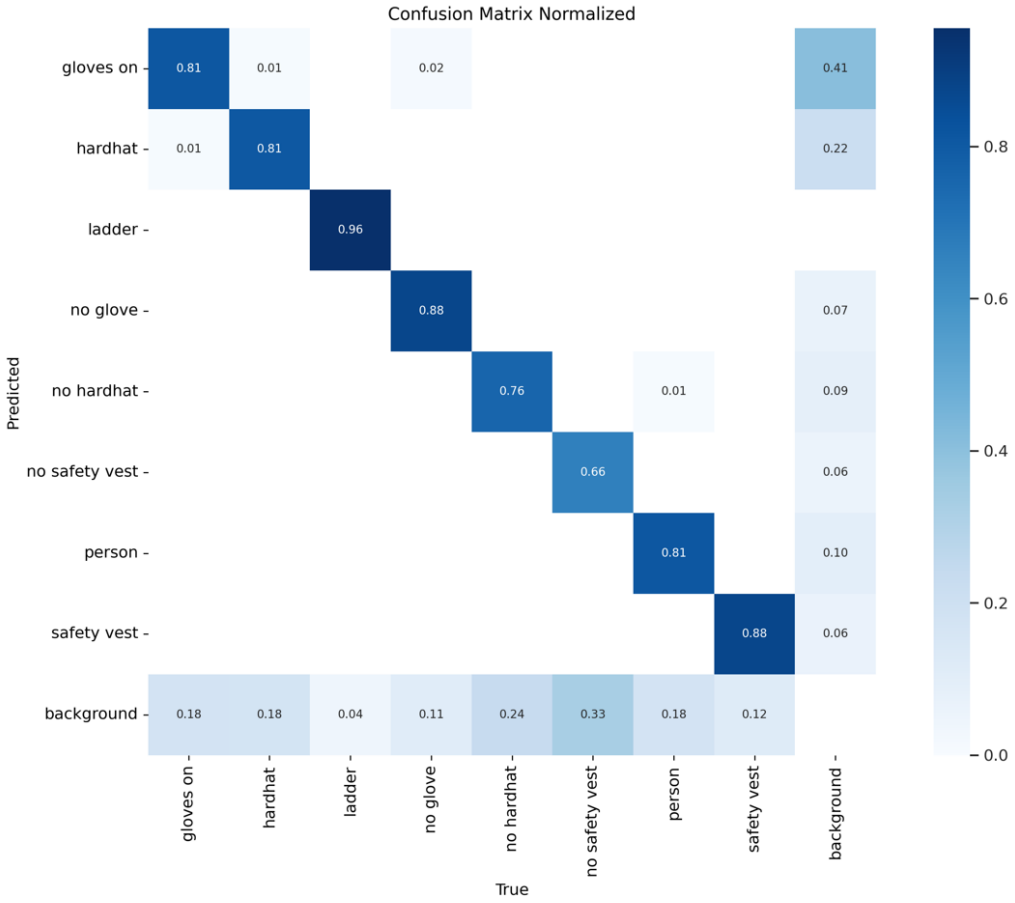


Figure 4. Normalized Confusion Matrix.

As observed from Figure 4, the model performed very well on the "ladder," "no glove," and "safety vest" classes, achieving normalized scores of 0.96, 0.88, and 0.88, respectively. Performance in other classes is more moderate. Notably, among all background instances, the model incorrectly predicted "no safety vest" 33% of the time and "no hardhat" 24% of the time. An example set of images from the validation set is shown in Figures 5 and 6 below. Figure 5 shows the ground-truth labels, and Figure 6 shows the trained model's predictions on the same set of images. Confidence scores of object detections are included

with the predictions. This comparison between Figures 5 and 6 visualizes the information shown in the Confusion matrix.

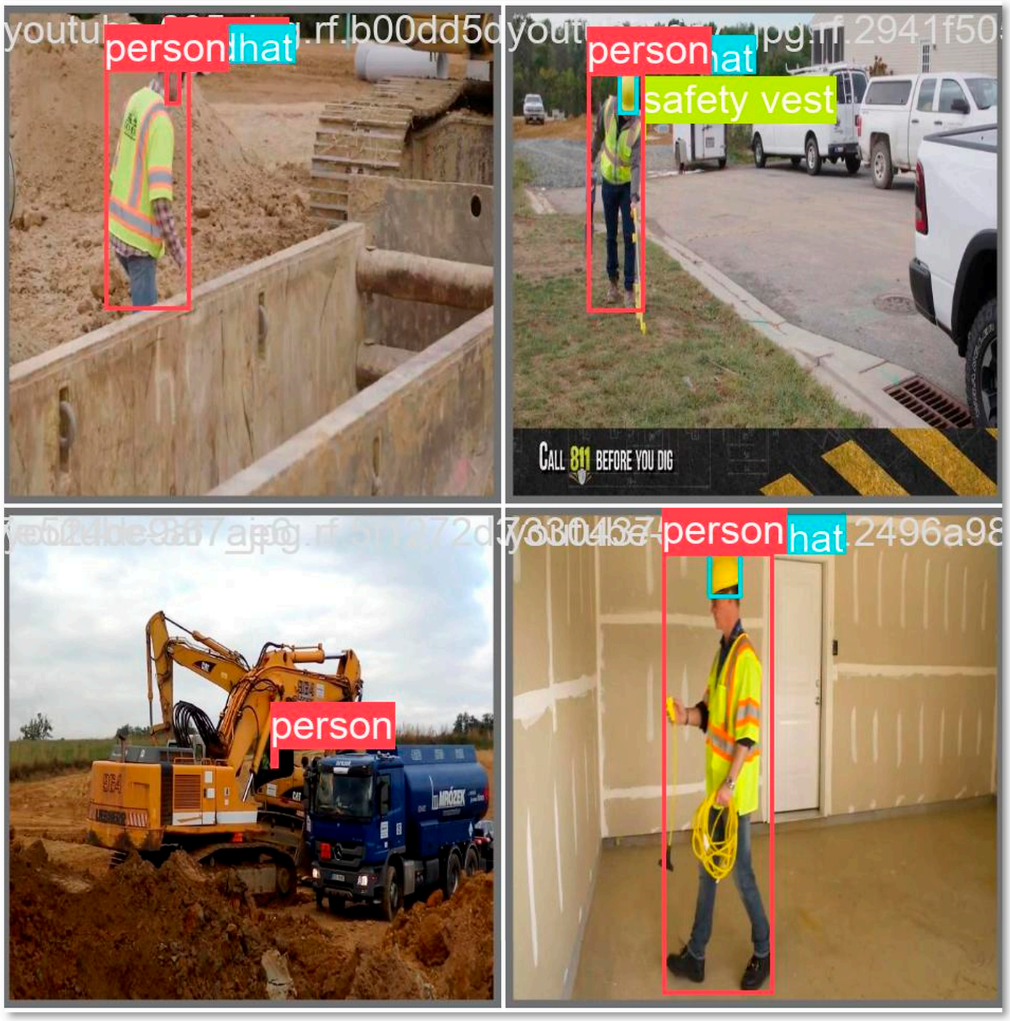


Figure 5. Validation Set example images: ground-truth labels.

Figure 7 below shows the trained model's detections on a test image produced in a lab setting.

The Precision–Confidence (P-C) Curve, shown in Figure 8, measures the correctness or quality of the positive predictions. Per Equation (2), precision is the proportion of predicted positives that are actually correct. Each thin line depicts how precision changes as the confidence threshold increases for each class, and the thick blue line shows the average across all classes. The "ladder" and "hardhat" achieve near-perfect precision by the 0.7 confidence threshold. The "gloves on" class again has the weakest performance, while "no safety vest" and "person" show more moderate precision.

An unusual dip is observed in the "no hardhat" class near a confidence threshold of 1.0, suggesting that the model made at least one highly confident but incorrect prediction. At such extreme thresholds, even a single false positive can significantly reduce the precision score due to the low number of predictions. This highlights a potential risk of overconfidence in ambiguous visual conditions, especially if an excessively high threshold is used in real-world deployment.

When evaluating a classifier using precision and recall, it is common to visualize the trade-off between the two with a Precision–Recall Curve [28]. Figure 9 shows the Precision–Recall Curve across the full range of confidence thresholds (0.0 to 1.0) for the

Worker Safety model. The legend on the right also displays the Mean Average Precision for each class at an IoU threshold of 0.5, quantifying the trade-off illustrated by the curves. The overall mAP across all classes is 0.827, indicating strong relative detection performance. This attribute is particularly important in the context of construction environments where negative or background samples vastly outnumber positive hazard instances.



Figure 6. Validation Set example images: trained model's predictions.



Figure 7. Test Set example image.

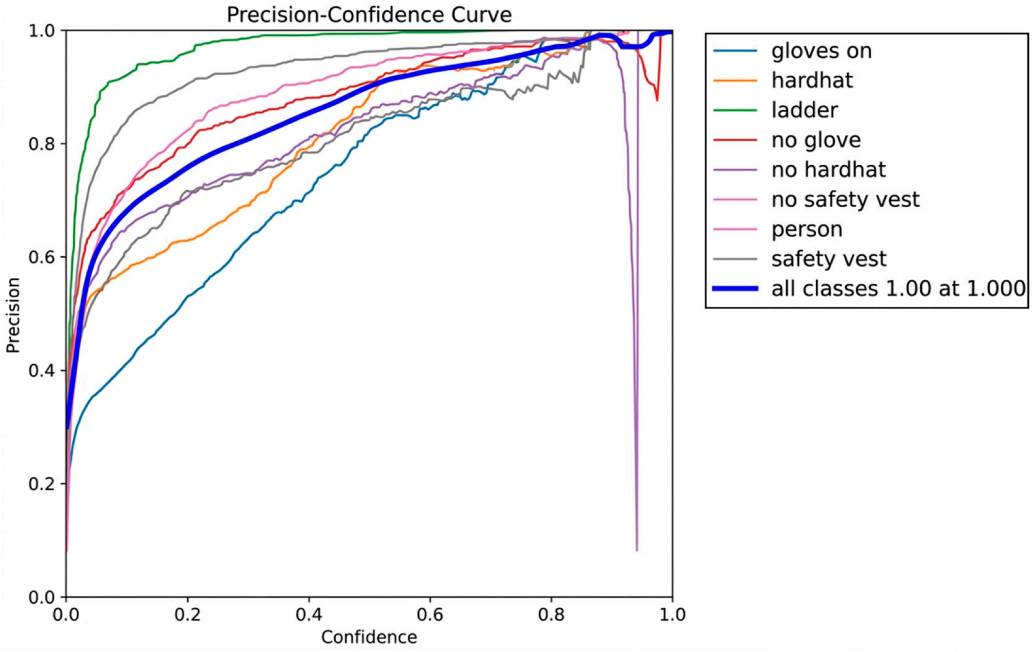


Figure 8. Precision–Confidence Curve.

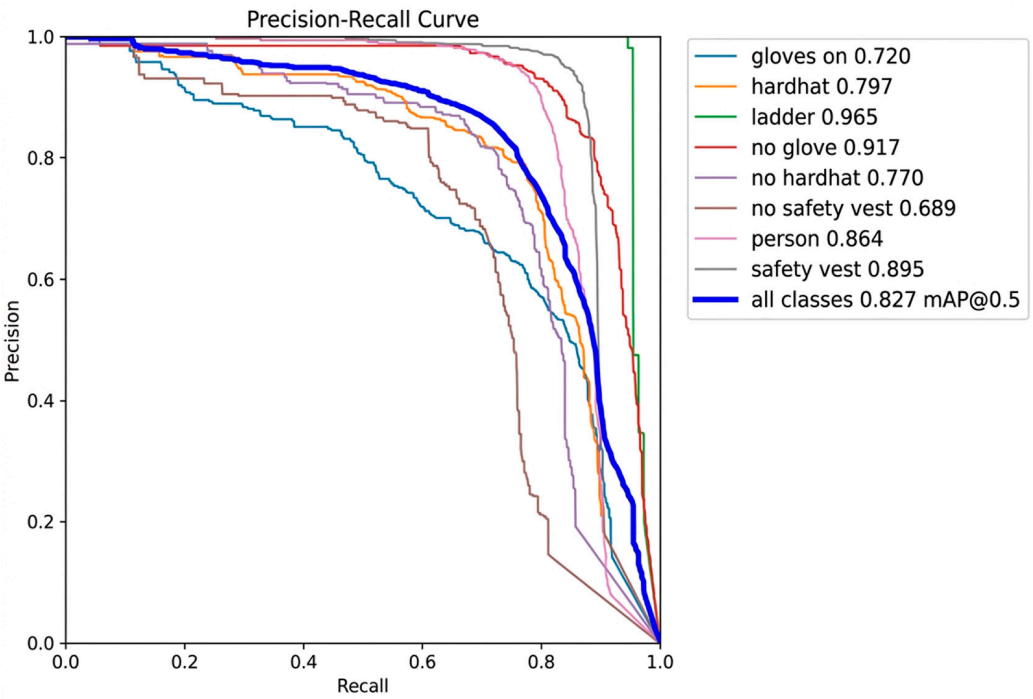


Figure 9. Precision–Recall Curve.

To distinguish model development results from final generalization performance, the trained YOLOv11 model was also evaluated on the independent test subset containing 660 images. Unlike the validation set, which was used during model development to assess training behavior and confidence threshold performance, the test subset was held out from training and used for final evaluation. Table 2 below compares the validation and independent test set results using the same performance metrics discussed previously. Overall mAP and class-specific AP values are reported at an IoU threshold of 0.5.

Table 2. Comparison between Validation and Test Set Results.

Metric	Validation Set	Test Set	Difference
Overall precision	0.841	0.830	−0.011
Overall recall	0.779	0.750	−0.029
Overall F1	0.809	0.788	−0.021
Overall mAP@0.5	0.827	0.763	−0.064
Overall mAP@0.5:0.95	0.614	0.545	−0.069
Gloves on AP@0.5	0.720	0.775	+0.055
Hardhat AP@0.5	0.798	0.811	+0.013
Ladder AP@0.5	0.965	0.949	−0.016
No glove AP@0.5	0.917	0.887	−0.030
No hardhat AP@0.5	0.770	0.617	−0.153
No safety vest AP@0.5	0.691	0.404	−0.287
Person AP@0.5	0.864	0.796	−0.068
Safety vest AP@0.5	0.894	0.865	−0.029

The independent test set produced an overall precision of 0.830, a recall of 0.750, an F1 score of 0.788, a mAP of 0.763, and a mAP averaged across multiple IoU thresholds from 0.50 to 0.95 of 0.545. These values are lower than the validation results, which provide a more conservative estimate of model generalizability. The largest decrease occurred for the no safety vest class, where AP decreased from 0.691 on the validation set to 0.404 on the independent test set. This result indicates that the safety vest remains the most difficult class to generalize across unseen images, potentially because it depends on distinguishing the absence of a visual PPE item rather than detecting a clearly visible object.

3.2. End-User Interface

The augmented reality interface, which is the digital overlay in the user's field-of-view and is also known as the "Heads-up Display", is shown in Figure 10. The display shows four construction zones, A, B, C, and D, which correspond to the location of the cameras set up on a representative construction site. Each zone displays the number of people in each construction zone along with any active PPE violations in that zone, with updates every 5 s. Users can close this menu by pressing the return button (i.e., through the AR interface). This AR interface provides a practical way for SSHOs to keep their head up and eyes watching the work performed in the construction zone in front of them while simultaneously being informed of critical safety information in other zones. This interface may support SSHO situational awareness by allowing safety-related information from other zones to be viewed while maintaining attention on nearby work activities.

The AR interface was intentionally designed to display zone-level alerts rather than continuous image overlays, video feeds, or bounding boxes in the SSHO's field of view. This design was selected to reduce visual clutter and allow the SSHO to maintain direct visual attention on nearby work activities. The interface is therefore intended to increase situational awareness by notifying the SSHO that a specific zone requires attention, rather than replacing direct field observation or providing precise visual localization of each detected worker. For this reason, the system is best suited for trained SSHOs who can interpret zone-based alerts, visit the indicated work area, and confirm PPE-related observations in person.

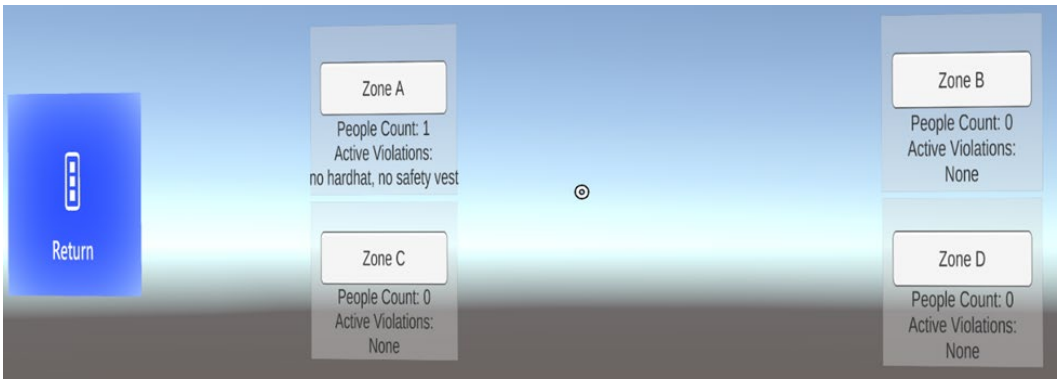


Figure 10. Augmented reality interface.

Figure 11 shows the web-based user interface designed for management personnel located in an office trailer or practically anywhere off-site. The interface displays the class name, confidence score, timestamp, people count, and active violations for Worker Safety Model detections across four construction zones, as well as detection information from the HoloLens 2 live stream, which updates every 5 s. The user also has the option to view the live stream from the HoloLens 2. Furthermore, the interface presents satellite imagery of each construction zone using the Leaflet API, based on the designated location of each zone. The map locations shown in Figure 10 are representative images chosen and are provided only for illustrative purposes. This interface allows the rest of the construction management team, such as the superintendent and the quality control manager, to have a snapshot of potential safety issues and to assist the SSHO in correcting PPE violations as needed. Off-site management, such as the project manager or the corporate safety manager, can also periodically monitor this safety information on the construction site as desired.

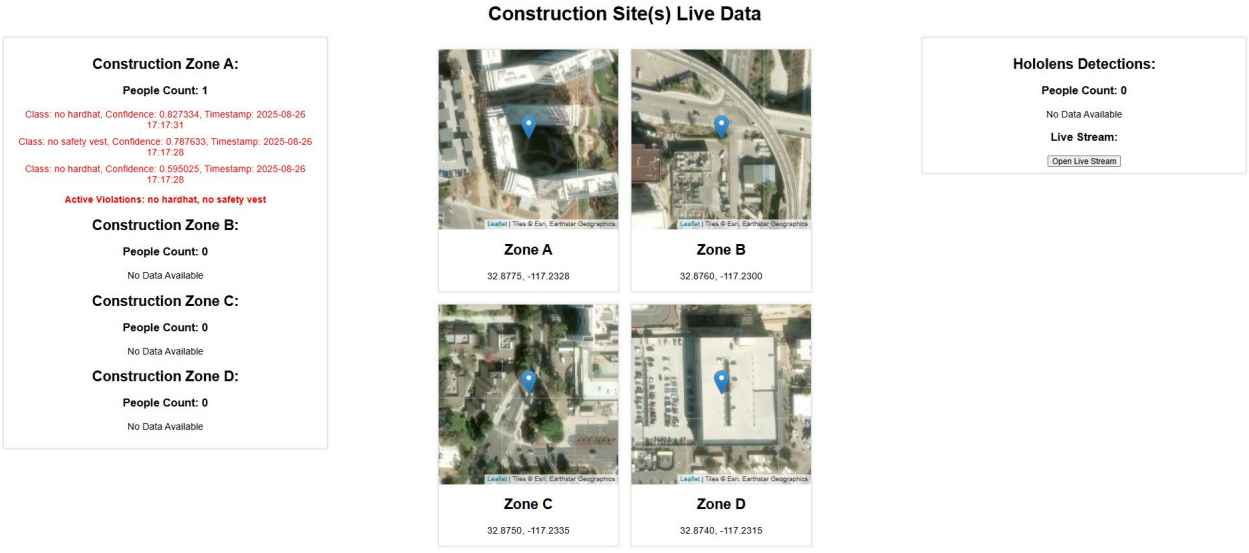


Figure 11. Web−based user interface.

The field validation confirmed that the integrated system could process two live camera feeds, detect workers and PPE compliance, update zone occupancy information, and display detection outputs on both the web interface and the HoloLens 2 heads-up display. Detection metadata, including class labels, confidence scores, timestamps, people counts, and active violations, was successfully transferred through the local XAMPP MySQL pipeline without requiring internet connectivity.

4. Discussion

4.1. Interpretation of Findings in Relation to the Existing Literature

The performance of this study's model varied across classes, with not all PPE classes detected equally well, but this is expected in real-world object detection tasks. Moderate to strong detection performance was noted for "person", "safety vest", "no safety vest", "hardhat", "no hardhat", "no glove", and "ladder" because of their relatively distinct shape and color with strong contrast against diverse backgrounds. However, the model sometimes demonstrated a propensity for false positives of "no safety vest" and "no hardhat", especially in visually ambiguous or low-resolution conditions. In real-world deployments, such tendencies may lead to false alarms when workers appear far from the camera or when image quality is degraded. "Gloves on" exhibited the weakest detection performance. This weak performance can be attributed to the small object size in combination with occlusion, non-distinct shape, and color contrast with diverse backgrounds. These model performance trends are consistent with known limitations for small and ambiguous objects presented in other recent studies [12–14]. These observations further emphasize the importance of interpreting detection outputs within the context of real-world deployment conditions on construction sites, where object size, occlusion, diverse backgrounds, and image quality can influence reliability.

Overall, the trained YOLOv11 model achieved a mAP of 0.827, which indicates strong detection performance across classes. This result is consistent with recent studies whose focus was to improve detection performance by making modifications or additions to the YOLO architecture [11–15]. While these prior works are model-centric and performed under controlled or benchmark conditions, the present study demonstrates comparable detection performance within a system-level deployment of a multi-component pipeline, which includes live camera feeds, database communication, and augmented reality visualization. Despite real-time deployment constraints, such as latency overhead and system integration in the field, this work shows that reliable and accurate PPE detection can be integrated with field data communication and visualization tools to provide safety-related information to construction management personnel under representative deployment conditions.

4.2. Limitations and Future Work

The current model of this Worker Safety system focuses on PPE detection, which provides a limited scope of identifying potential hazards. Movement tracking and gait analysis will be developed and included in the system in future models to identify potentially injured or fatigued workers.

The field validation was limited in duration and scale and was designed as a proof of concept rather than a controlled operational trial. Although the test verified zone occupancy detection, PPE compliance detection, and data pipeline functionality on an outdoor construction site with seven workers, it did not include a formal measurement of incident reduction, SSHO response time, or statistically sampled false alarm rates. Future work should include longer field deployments with systematic logging of detections, false positives, false negatives, alert timing, and user response times. Also, the Microsoft HoloLens 2, along with many other commercially available AR headsets, is currently too bulky for practical wear on the construction site. Outdoor visibility is also a significant limitation for construction applications. During field testing, the Microsoft HoloLens 2 rendered the interface clearly under most lighting conditions, but the menu could become difficult to see or invisible in direct bright sunlight. This limitation is important because many construction activities occur outdoors, and SSHOs may need to use the interface while moving between exposed work areas. Future work should evaluate display brightness, contrast settings, and alternative AR hardware designed for outdoor use. The implementation of this AR

system on the HoloLens 2 provides a proof of concept that can be built upon in the future. As AR glasses decrease in size, this AR system will become more practical.

The generalizability of the object detection model used for this system may be influenced by different colors, shapes, or standards of PPE in other countries. The system was developed for United States federal construction projects and is meant to detect common PPE required on these construction sites. The object detection model can be fine-tuned for PPE used in other countries, or a different object detection model can be used. The modularity of this system allows for different models to be deployed with minimal impact to the rest of the system or data flow, with minor edits required to the data handling scripts only if the class names are changed.

4.3. Ethical Considerations

Privacy considerations are at the forefront of critical feedback for this project from those in the construction industry. The detection metadata stored in the local database is limited to safety-related detection outputs, including class labels, confidence scores, bounding box coordinates, timestamps, construction zones, and aggregate person counts. The system does not store video streams, perform facial recognition, collect biometric identifiers, or associate detections with worker names or personnel records. Therefore, this system is intended as a safety tool and not a tool for monitoring crew productivity. Future deployments should include access controls, data retention limits, and policies restricting the use of detection records to safety monitoring.

5. Conclusions

This research showcased the integration of real-time ML and AR for safety interventions for construction site safety monitoring. A modular system was developed that detects the presence of PPE across multiple construction zones simultaneously and delivers actionable hazard alerts through both a web-based interface and a wearable heads-up display. The YOLOv11 model achieved strong performance, with an overall mAP of 0.827, and was field-tested in a realistic environment with accessible, low-cost hardware. These results establish the potential for a practical framework for site-wide safety awareness, while future work should evaluate user response time, usability, workflow integration, broader hazard detection, and a lighter AR platform for industry adoption.

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Data Availability Statement: The initial dataset used in this study was developed by the authors on Roboflow Universe and is available at https://app.roboflow.com/armor/workersafety25/2 (accessed on 08 June 2026). Before model training, the exported YOLO dataset configuration file was inspected to verify consistency with the annotation set. The export included an unused, undefined class entry, although no images and annotations corresponded to that class. This unused entry was removed so that the training configuration matched the eight object classes reported in Table 1. The cleaned configuration file and exact dataset version used for model training and evaluation are available

via https://osf.io/tyd5m/overview?view_only=9832b6e8c6cc4f71b89dd951e2784dea (accessed on 8 June 2026).

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Abbreviations

The following abbreviations are used in this manuscript:

AR	Augmented Reality
BLS	Bureau of Labor Statistics
CNN	Convolutional Neural Network
CV	Computer Vision
CCTV	Closed-Circuit Television
EM-385-1-1	U.S. Army Corps of Engineers Safety and Health Requirements Manual
FN	False Negative
FP	False Positive
HAR	Human Activity Recognition
HTTP	Hypertext Transfer Protocol
IoU	Intersection Over Union
JSON	JavaScript Object Notation
mAP	Mean Average Precision
AP	Average Precision
ML	Machine Learning
OSHA	Occupational Safety and Health Administration
PPE	Personal Protective Equipment
QCM	Quality Control Manager
R-CNN	Region-based Convolutional Neural Network
RTSP	Real-time Streaming Protocol
SSHO	Site Safety and Health Officer
TP	True Positive
UI	User Interface
YOLO	You Only Look Once

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